

PAPER 1 / QUESTION 106

5 November 2025 / TZ3 (file label) / HL Paper 1A / original question 5 / 8825-6719

- 106.** A mass m is attached to a string and moves in a vertical circle of constant radius R . At the top of the circle, the tension in the string is T . Air resistance is negligible.

What is the kinetic energy of m at the top of the circle?

- A. $\frac{R(T - mg)}{2}$
- B. $\frac{R(T + mg)}{2}$
- C. $R(T - mg)$
- D. $R(T + mg)$